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Noise muffler for microphone

Abstract

A noise muffler for a microphone includes a circumferential sidewall, a shroud, a hollow support stem, and a sponge. The circumferential sidewall extends from a rear wall and includes a curved inner surface. The shroud extends from at least a portion of the circumferential sidewall and comprises a curved outer surface and a curved inner surface. The hollow support stem extends from a generally central location of the rear wall and is aligned with an opening in the rear wall. The support stem includes a plurality of openings uniformly positioned around an outer surface of the support stem. The sponge is disposed around the support stem and conforms with the curved inner surface of the shroud and the curved inner surface of the sidewall. The shroud and the sponge deflect airflow away from the opening in the rear wall.

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Background/Summary

BACKGROUND

(1) Microphones are used in various environments, including outdoor environments with varying wind conditions. In addition to capturing the intended sound (e.g., a person speaking), microphones capture noises caused by external factors such as wind generated turbulence. The noise caused by external factors interferes with the quality of the intended sound being captured by the microphone, affecting the quality of the sound profile captured by the microphone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Throughout the drawings, reference numbers may be re-used to indicate correspondence between referenced elements. The drawings are provided to illustrate example embodiments described herein and are not intended to limit the scope of the disclosure.

(2) FIG. **1**A is a front view of an object having a noise muffler for a microphone;

(3) FIG. **1**B is a side view of the object of FIG. **1**A;

(4) FIG. **2** is a perspective view of a noise muffler for a microphone;

(5) FIGS. **3** and **4** are perspective views of the noise muffler of FIG. **2** with a sponge removed for illustrative purposes;

(6) FIGS. **5** and **6** are side views of the noise muffler of FIG. **2** with the sponge removed for illustrative purposes;

(7) FIG. **7** is a front view of the noise muffler of FIG. **2** with the sponge removed for illustrative purposes;

(8) FIG. **8** is a rear view of the noise muffler of FIG. **2** with the sponge removed for illustrative purposes;

(9) FIG. **9** is a top view of the noise muffler of FIG. **2**;

(10) FIG. **10** is a bottom view of the noise muffler of FIG. **2** with the sponge removed for illustrative purposes; and

(11) FIG. **11** is a cross-sectional view of the noise muffler of FIG. **2** with the sponge removed for

illustrative purposes.

DETAILED DESCRIPTION

(12) The present disclosure generally relates to noise mufflers for use with microphones, specifically microphones used in outdoor environments where air flow, such as wind, can interfere with the intended sound (e.g., a person speaking) being captured by the microphone. Air flow caused by wind can be unpredictable and constantly varying. The direction of air flow can constantly change, for example, air flow can be left to right and then switch to right to left or any other direction. As air flows over non-aerodynamic surfaces, turbulence is generated which in turn generates noise that is collected as part of the sound wave profile. Proximity of the turbulence to a microphone can drastically affect the sound profile. When air flow interferes with the sound being captured by a microphone, the microphone may no longer work for its intended purpose. Therefore, it is desirable to have noise mufflers capable of reducing, limiting, or preventing noises from outside sources, such as air flow and wind, to be captured by the microphone.

(13) The present disclosure relates to aerodynamic deflection based on air flow characteristics over a generally cylindrical, generally spherical, and/or aerodynamic body. The generally cylindrical, generally spherical, and/or aerodynamic body can reduce and/or eliminate the turbulence that leads to wind noise perturbations. In some instances, the generally cylindrical, generally spherical, and/or aerodynamic body can cause the turbulence to occur farther downfield away from the sound receiving microphone, allowing for clearer capture of the desired sound by the microphone. Additionally, the noise mufflers according to the present disclosure provide for an amplification effect from an inside curvature that is capable of concentrating the intended sound waves towards the receiving microphone.

(14) The noise mufflers according to the present disclosure may be used with exterior microphones, such as microphones in outdoor environments that experience air flow caused by wind. One specific non-limiting example includes microphones at entrance and/or exit gates of a facility that are used to communicate with drivers of vehicles attempting to enter and/or exit the gates. A remote agent may rely on the use of a microphone to communicate with the driver of the vehicle and therefore there is a need to reduce or eliminate the turbulence caused by wind the microphone may be exposed to. If the turbulence is not reduced and/or eliminated, a person located at the facility may need to stop what they are doing to travel to the entrance and/or exit gate to communicate with the driver in person. Additional, non-limiting examples of use include microphones at drive through restaurants, microphones mounted on vehicles, and microphones used in outdoor environments (e.g., outside buildings). Therefore, it is desirable to implement noise mufflers according to the present disclosure to improve the quality of the sound captured by exterior microphones.

(15) FIGS. **1**A-**11** show various views of a noise muffler **100**. FIGS. **1**A and **1**B are front and side views of the noise muffler **100** mounted to an object **104**. The object **104** can be any object that includes a microphone, for example a communication box at an entrance and exit gate that includes a microphone to communicate with a driver of a vehicle attempting to enter and/or exit the gate. The object **104** may be any object **104** that experiences air flow or unexpected wind that may interfere with the sound that is intended to be identified by the microphone. As shown in FIG. **1**B, the noise muffler **100** can extend away from a front surface **105** of the object **104**, allowing air flow to flow over, under, and/or around the noise muffler **100**.

(16) FIGS. **2-4** are perspective views of the noise muffler **100**. The noise muffler **100** can include a rear wall **108**, a sidewall **112** extending from the rear wall **108**, a shroud **116** and a support stem **134**. A microphone **102** can in one example be positioned parallel to the rear wall **108**, for example as labeled in FIG. **8**. In some embodiments, the microphone **102** can be adjacent and/or next to the noise muffler **100**. In some embodiments, the microphone **102** can be flush with the rear wall **108** of the noise muffler **100**. In some embodiments, the microphone can be positioned linearly offset from the opening **103**. In some embodiments, the microphone can be positioned directly behind or

aligned with the opening **103**. The rear wall **108** can couple or attach the noise muffler **100** to the object **104**. In some embodiments, the rear wall **108** can have a generally circular cross-section. In some embodiments, the sidewall **112** can have a generally circular cross-section (e.g., can extend circumferentially about an axis of the noise muffler **100**). The sidewall **112** and the rear wall **108** can be connected via a curved transition wall **113**, for example as shown in FIGS. **5** and **6** illustrating side views of the noise muffler **100**. In some embodiments, the curved transition wall **113** can be generally U-shaped. The curved transition wall **113** can decrease in diameter from the rear wall **108** to the start of the sidewall **112**. The diameter of the sidewall **112** can gradually increase from the curved transition wall **113** away from the rear wall **108**. In some embodiments, the sidewall **112** and the rear wall **108** can be connected without the curved transition wall **113**. For example, the sidewall **112** may extend from the shroud **116** and continue on a curved path away from a central axis A**1** (as labeled in FIG. **11**) to the rear wall **108**.

(17) The sidewall **112** can have an inner surface **114**, for example as shown in FIGS. **3**, **4**, and **11**. The inner surface **114** can advantageously be shaped to direct sound waves toward the microphone **102**. In one example, the inner surface **114** can have a curvature, such as a spherical curvature, which directs sound toward the microphone **102**. In another example, the inner surface **114** can have a conical shape that directs sound toward the microphone **102**. The inner surface **114** can extend from an outer end of the sidewall **112** to the rear wall **108**. In some embodiments, the inner surface **114** can extend from the outer end of the sidewall **112** to a location **115** (as labeled in FIG. **11**) where the support stem **134** extends from the rear wall **108**. In some embodiments, a front surface **110** of the rear wall **108** can be generally flat, for example, as shown in FIG. **11**. In some embodiments, the front surface **110** of the rear wall **108** can be curved, for example, the inner surface **114** of the sidewall **112** can have the same curvature as the front surface **110** of the rear wall **108**, forming a generally semi-spherical shape. The inner surface **114** can advantageously create an amplification effect that concentrates sound waves towards the microphone **102**. In some embodiments, the sidewall **112** can include a curved, rounded, or tapered front end **121**.

(18) The noise muffler **100** can include the shroud **116** extending from the sidewall **112**. The shroud **116** can include a first portion **120** and a second portion **124**. The first portion **120** of the shroud **116** can extend at a first end from the sidewall **112**. In some embodiments, the first portion **120** of the shroud **116** can extend from a portion of the sidewall **112** positioned above horizontal axis **117**, as labeled in FIG. **7**. In some embodiments, the first portion **120** of the shroud **116** can extend from a portion of the sidewall **112** and have a generally semi-circular cross-section. The first portion **120** of the shroud **116** can have a curved outer surface **128** and a curved inner surface **118** forming a generally semi-cylindrical shape. The first portion **120** can be generally C-shaped. The first portion **120** can have an arch shape. In some embodiments, end walls **122** of the first portion **120** of the shroud can be curved, rounded, or tapered.

(19) The second portion **124** of the shroud **116** can extend from a second end of the first portion **120** of the shroud **116**. The second portion **124** of the shroud **116** can have a generally quarter spherical shape. For example, the second portion **124** can have an outer curved surface **129** (e.g., defined by a spherical curvature) extending in the X-direction, as labeled in FIG. **5**, and in the Y-direction, as labeled with reference to FIG. **7**. The outer curved surface **129** can align (e.g., tangentially) with the outer surface **128** of the first portion **120** to provide a continuous outer surface for the shroud. In one example, the curved outer surface **129** can be a spherical curvature. The curved outer surface **129** can extend toward the horizontal axis **117**, as labeled in FIG. **7**. The second portion **124** of the shroud **116** can include a curved inner surface **119**. In some embodiments, end walls **123** of the second portion **124** of the shroud can be curved, rounded, or tapered.

(20) The curved transition wall **113**, the sidewall **112**, and the shroud **116** when viewed together can define a generally spherical or spheroid shape. In some embodiments, the curved transition wall **113**, the sidewall **112**, and the shroud **116** when viewed together can form a capsule shape having

two semi-spherical ends and a semi-cylindrical or cylindrical body in between the two semi-spherical ends.
(21) In some embodiments, the second portion **124** can have a cut-out **126** in a forward facing surface of the shroud **116**, for example, as shown in FIGS. **2-4** and **7**. The cut-out **126** can be defined by an edge **127**, as shown in FIG. **5**. The edge **127** can extend along a plane. The cut-out **126** can be generally semi-circular in shape. The cut-out **126** can allow for straight sound transmission to a receiving microphone. In some embodiments, the edge **127** of the cut-out **126** can be sharp (e.g., not rounded or curved). The sharp edge **127** can prevent rain, snow, or other precipitation from entering the noise muffler by directing the rain, snow, or other precipitation downward so that it falls to the ground.
(22) The noise muffler **100** can include a support stem **134** extending from the front surface of the rear wall **108**, for example as shown in FIGS. **5**, **6**, and **11**. The support stem **134** can extend a distance D**1** (as labeled in FIG. **11**). The distance D**1** can be less than a distance from the from the front surface **110** of the rear wall **108** to the edge **127** of the cutout **126**. In some embodiments, there may be a distance D**2** (as labeled in FIG. **11**) defined by an end of the support stem **134** and an inner wall of the shroud **116**. An outer surface of the support stem **134** and inner surfaces **114**, **118**, **119** may define a distance D**3**. The distance D**3** may vary along the distance D**1** (e.g., the length) of the support stem **134**. The distances D**2** and D**3** may allow room for a sponge to be disposed over the support stem **134**, as described in more detail below.
(23) The support stem **134** can be aligned with an opening **130**. The opening **130** can extend from a rear surface **111** to a front surface **110** of the rear wall **108**, for example as shown in FIGS. **7**, **8**, and **11**. The opening **130** can be positioned in a generally central location of the rear wall **108** in an area surrounded by the perimeter of the sidewall **112**. In some embodiments, the opening **130** can be circular in shape. In some embodiments, a diameter D**4** of the opening **130** may increase from the front surface **110** of the rear wall **108** to a rear surface **111** of the rear wall **108**, as shown in FIG. **11**. In some embodiments, the opening **130** can be conical or tapered (e.g., to allow sufficient room for the microphone so that rear wall **108** does not obstruct or cover the microphone). In other embodiments, the opening **130** can be cylindrical (e.g., have a constant diameter D**4**). The shape of the opening **130** (e.g., conical, tapered, cylindrical) can help direct the intended sound toward the microphone (e.g., microphone **102**). For example, in embodiments where the microphone is positioned linearly offset from the opening **130**, the opening **130** being tapered can assist in directing the intended sound toward the microphone. In embodiments, where the microphone is positioned behind or aligned with the opening **130**, the opening **130** being cylindrical can assist in directing the intended sound toward the microphone.
(24) The support stem **134** can be hollow and form a channel that is aligned with the opening **130**. The support stem **134** can be centrally located on the rear wall **108**. In some embodiments, the support stem **134** can include a plurality of openings **138** (see FIGS. **5-6**, **10**, **11**) in communication (e.g., fluid communication) with the channel of the support stem **134** and in communication with the opening **130** (e.g., so that air or sound can flow through the openings **138** into the channel of the support stem **134** and into the opening **130**). The plurality of openings **138** can be uniformly spaced circumferentially and linearly along the support stem **134**. In some embodiments, the plurality of openings **138** can form one or more rows of openings **138** extending along the length of the support stem **134**. In some embodiments, the plurality of openings **138** can form one or more rings of openings **138** extending around the perimeter of the support stem **134**. The plurality of openings **138** extend from an outer surface of the support stem **134** to an inner surface of the support stem **134**. In some embodiments, the openings **138** may be generally rectangular in shape, although other shapes are possible.
(25) The plurality of openings **138** can form connectors **139** that form the support stem **134**. The connectors **139** can be thin structures that define the support stem **134** walls. The size and shape of the connectors **139** can be dependent upon the size and shape of the openings **138**. In some

embodiments, the connectors **139** may be generally linear. In some embodiments, at least one connector **139** may extend generally parallel to a central axis A**1** (as labeled in FIG. **11**) of the noise muffler **100**. In some embodiments, at least one connector **139** may extend generally perpendicular to the central axis A**1** of the noise muffler **100**. The combination of the plurality of openings **138** and the connectors **139** can increase the structural strength of the support stem **134** and prevent or inhibit damage to the support stem **134**.

(26) In some embodiments, a sponge **140**, as shown in FIGS. **1**A, **1**B and **2**, can be disposed over the support stem **134**. The sponge **140** can be made of foam or other suitable sponge material (e.g., a material that is compressible, resilient and has favorable acoustic properties). The general central location of the support stem **134** on the rear wall **108** can assist in centralizing the sponge **140** and assist in maintaining the positioning of the sponge **140** (e.g., within the shroud **116**). Additionally, the support stem **134**, the shroud **116**, and/or the sidewall **112** can function together to prevent or reduce the chance that the sponge **140** is blown away by wind in an outdoor environment. For example, the location of the stem **134** on the rear wall **108**, the spacing of the stem **134** from the shroud **116**, and the size of the sponge **140** can facilitate retaining the sponge **140** between the shroud **116** and the stem **134**. The sponge **140** can have a generally cylindrical shape.

(27) The sponge **140** can be generally spherical or conical in shape. The sponge **140** can correspond to the curvature of the inner surface **114** of the sidewall **112** and the curved inner surfaces **118**, **119** of the shroud **116**. In some embodiments, the sponge **140** can conform to the general shape of the inner surface **114** of the sidewall **112** and the curved inner surfaces **118**, **119** of the shroud **116**. An outer surface of the sponge **140** can come in contact with the inner surface **114** of the sidewall **112** and the curved inner surfaces **118**, **119** of the shroud **116** and can also come in contact with an outer surface of the support stem **134**. In one example, the sponge **140** has a thickness (e.g., an uncompressed thickness) approximately equal to the distance D**3**. In another example, the sponge **140** has a thickness (e.g., an uncompressed thickness) greater than the distance D**3**, so that the sponge **140** is compressed between the support stem **134** and the curved inner surfaces **118**, **119** when disposed over the support stem **134**. The thickness of the sponge **140** and/or the contact between the sponge **140** and the inner surfaces **114**, **118**, **119** can assist in maintaining the shape of the sponge **140** and in retaining the sponge **140** on the stem **134** to inhibit or prevent the sponge **140** from being blown off the stem **134** by wind. In some embodiments, the shroud **116**, the sidewall **112**, and the sponge **140** can function as a single cylindrical, spherical, and/or aerodynamic structure. For example, the curvature of the shroud **116**, sidewall **112**, and sponge **140** can essentially form a spherical, cylindrical, and/or aerodynamic shape when assembled together. This can allow the air flow to be directed away from the receiving microphone (e.g., microphone **102**) and thus reduce or prevent any turbulence generated noise from being captured by the receiving microphone.

(28) In some embodiments, the thickness T**1** (as labeled in FIG. **11**) of the shroud **116** may be minimized such that the edge **127** and the outer surface of the sponge **140** are almost continuous. The thickness T**1** being minimized can allow for the edge **127** of the shroud **116** and the outer surface of the sponge **140** to be approximately tangential. Minimizing the distances D**2**, D**3** can provide for minimal space within an area defined by the shroud **116** and sidewall **112**. In some embodiments, it can be beneficial to minimize the thickness T**1** and/or the distances D**2**, D**3** to improve the quality of the intended sound traveling to the microphone (e.g., microphone **102**) by deflecting airflow over the shroud **116** and sponge **140** to minimize turbulent air flow near the noise muffler **100** that can affect (e.g., reduce) the quality of the sound captured by the microphone.

(29) The shroud **116** can protect the sponge **140** from water damage due to rain, snow, or other weather conditions. The shroud **116** can cover a top surface of the sponge **140**. The shroud **116** can at least partially cover side surfaces of the sponge **140**. The shroud **116** can at least partially cover a front surface of the sponge **140**. The sponge **140** can be removable and/or replaceable from the noise muffler **100**.

(30) In some embodiments, the noise muffler **100** can include one or more recesses **144** configured to mount the noise muffler **100** to the object **104**, for example, as shown in FIG. **8**. The recesses **144** can be sized and shaped to mount the noise muffler **100** to protrusions extending from the object **104**.

(31) During use, the noise muffler **100** can deflect and/or redirect airflow perturbations. For example, the cylindrical, spherical, and/or aerodynamic shape of the noise muffler **100** can redirect the airflow away from the receiving microphone (e.g., the microphone **102**). For example, using a left to right air flow as a non-limiting example, arrow **150** in FIG. **7** illustrates how air flow can approach the noise muffler **100**, engage the shroud **116**, travel over the shroud **116** and away from the receiving microphone. Another non-limiting example of air flow traveling towards the front end of the noise muffler **100** is illustrated by arrow **151** in FIG. **5**. As shown, air flow can approach the noise muffler **100** from the front, engage the shroud **116**, travel over the shroud **116** and away from the receiving microphone. The sponge **140** can function in a similar way as the shroud **116** in directing air flow away from the receiving microphone. For example, as shown by arrows **152** and **153** in FIG. **2**, air flow approaching the sponge **140**, from either the front or side can be directed away from the receiving microphone.

(32) While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

(33) Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

(34) Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

(35) Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may

be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

(36) For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

(37) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

(38) Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(39) Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately”, “about”, “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

(40) The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

(41) Of course, the foregoing description is that of certain features, aspects and advantages of the present invention, to which various changes and modifications can be made without departing from the spirit and scope of the present invention. Moreover, the devices described herein need not feature all of the objects, advantages, features and aspects discussed above. Thus, for example, those of skill in the art will recognize that the invention can be embodied or carried out in a manner that achieves or optimizes one advantage or a group of advantages as taught herein without necessarily achieving other objects or advantages as may be taught or suggested herein. In addition, while a number of variations of the invention have been shown and described in detail, other modifications and methods of use, which are within the scope of this invention, will be readily apparent to those of skill in the art based upon this disclosure. It is contemplated that various combinations or subcombinations of these specific features and aspects of embodiments may be

made and still fall within the scope of the invention. Accordingly, it should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the discussed devices.

Claims

1. A noise muffler for a microphone comprising: a circumferential sidewall extending from a rear wall, the circumferential sidewall comprising a curved inner surface; a shroud extending from at least a portion of the circumferential sidewall, the shroud comprising a curved outer surface and a curved inner surface; and a sponge disposed around a support stem extending from the rear wall, wherein the sponge conforms with the curved inner surface of the shroud and the curved inner surface of the circumferential sidewall, wherein the shroud is configured to deflect airflow away from a microphone.
2. The noise muffler of claim 1, wherein the sponge is removable from the support stem.
3. The noise muffler of claim 1, further comprising a hollow support stem extending from the rear wall, the hollow support stem comprising a plurality of openings uniformly positioned around an outer surface of the hollow support stem.
4. The noise muffler of claim 1, further comprising the microphone, wherein the microphone is positioned parallel with the rear wall.
5. The noise muffler of claim 1, wherein a front end of the shroud comprises a cut-out.
6. The noise muffler of claim 1, wherein the shroud has a semi-circular cross-sectional shape.
7. A noise muffler for a microphone comprising: a circumferential sidewall extending from a rear wall, the circumferential sidewall comprising a curved inner surface; a shroud extending from at least a portion of the circumferential sidewall, the shroud comprising a curved outer surface and a curved inner surface; and a hollow support stem extending from the rear wall, the hollow support stem comprising a plurality of openings uniformly positioned around an outer surface of the hollow support stem, wherein the shroud is configured to deflect airflow away from a microphone.
8. The noise muffler of claim 7, further comprising the microphone, wherein the microphone is positioned parallel with the rear wall.
9. The noise muffler of claim 7, wherein a front end of the shroud comprises a cut-out.
10. The noise muffler of claim 7, wherein the shroud has a semi-circular cross-sectional shape.
11. The noise muffler of claim 7, further comprising a sponge disposed around a support stem extending from the rear wall.
12. The noise muffler of claim 11, wherein the sponge is removable from the support stem.
13. A noise muffler for a microphone comprising: a circumferential sidewall extending from a rear wall, the circumferential sidewall comprising a curved inner surface; and a shroud extending from at least a portion of the circumferential sidewall, the shroud comprising a curved outer surface and a curved inner surface, the curved outer surface and the curved inner surface defining a C-shaped cross-section; wherein the shroud is configured to deflect airflow away from a microphone.
14. The noise muffler of claim 13, further comprising a sponge at least partially surrounded by the shroud.
15. The noise muffler of claim 14, wherein the sponge conforms with the curved inner surface of the shroud and the curved inner surface of the circumferential sidewall.
16. The noise muffler of claim 14, further comprising a support stem extending from the rear wall, wherein the sponge is removable from the support stem.
17. The noise muffler of claim 13, further comprising a hollow support stem extending from the rear wall, the hollow support stem comprising a plurality of openings uniformly positioned around an outer surface of the hollow support stem.
18. The noise muffler of claim 13, further comprising the microphone, wherein the microphone is positioned parallel with the rear wall.

19. The noise muffler of claim 13, wherein a front end of the shroud comprises a cut-out.

20. The noise muffler of claim 13, wherein the shroud has a semi-circular cross-sectional shape.